

UNIT-III**SUPER CONDUCTIVITY:**

Zero resistance state, Meissner effect, Critical temperature, Critical current – Derivation for a cylindrical wire, Critical field - Type-I and type-II superconductors, Vortex state, Explanation for upper critical field.

Formation of Cooper pairs - Mediation of phonons, BCS Theory -Phase coherent state, Limitations – examples of systems with low and high electron-phonon coupling, Two-fluid model, Cooper pair Tunneling (Andreev reflection), Josephson junction, Flux quantization, DC and AC SQUID, Numerical Problems

Introduction:

Superconductors are materials that exhibit the state of superconductivity, in which they offer zero electrical resistance when dc current is passed through them, and they do not permit magnetic fields to penetrate them. As a result, an electric current flowing through a superconductor can continue to flow indefinitely without any energy loss or degradation.

Superconductivity was discovered on April 8, 1911, by Heike Kamerlingh Onnes. At the temperature of 4.2K, he observed that the resistance abruptly disappeared. In subsequent decades, superconductivity was observed in several other materials. In 1913, lead was found to be superconductor at 7 K, and in 1941 niobium nitride was found to superconduct at 16K. Before 1980, the critical temperature found cannot be higher than 30K. Those superconductors are called conventional superconductors. Theoretically, if these materials do not possess any disorder, then superconductivity can be fully explained by BCS theory for 3-dimensional samples. Also, the behaviour close to the superconducting transition was well described by the phenomenological Landau-Ginzburg theory.

A perfect superconductor is a material that exhibits two characteristic properties, namely zero electrical resistance and perfect diamagnetism, when it is cooled below a particular temperature T_c , called the critical temperature. At higher temperatures it is a normal metal and ordinarily is not a very good conductor. For example, lead, tantalum, and tin become superconductors, while copper, silver, and gold, which are much better conductors, do not superconduct. In the normal state some superconducting metals are weakly diamagnetic and some are paramagnetic. Below T_c they exhibit infinite electrical conductivity and perfect diamagnetism. Perfect diamagnetism, the second characteristic property, means that a superconducting material does not permit an externally applied magnetic field to penetrate its interior.

Superconductivity has a wide range of applications in various fields of science and technology. Here are some of the applications of superconductivity:

1. **Magnetic Resonance Imaging (MRI):** Superconducting magnets are used in MRI machines to create strong magnetic fields. These magnets produce a more accurate and detailed image of internal body organs and tissues, making MRI an essential tool in medical diagnosis.
2. **Particle Accelerators:** Superconducting magnets are used in particle accelerators to produce high-energy particle beams. This is important in high-energy physics research and can help us understand the fundamental properties of matter.
3. **Power Transmission:** Superconducting wires can transmit electricity with zero resistance, which means less energy is lost during transmission. This makes superconducting power cables more efficient and cost-effective compared to traditional power cables.
4. **Levitating Trains:** Superconductors can be used to levitate trains above the track, reducing friction and increasing efficiency.
5. **Energy Storage:** Superconducting energy storage devices can store large amounts of energy for long periods without any significant loss. This technology is important for renewable energy systems, where energy storage is crucial to managing fluctuations in supply and demand.
6. **Quantum Computing:** Superconducting materials are used in quantum computers to create qubits, which are the basic building blocks of quantum computers. These computers have the potential to perform calculations much faster than traditional computers.

The electrical conductivity of metals arises from the presence of free electrons in their outermost shells that can move freely in response to an electric field. When a current flows through a metal, these free electrons move opposite to the applied electric field. Meanwhile, the lattice ions, which are positively charged, vibrate around their equilibrium positions. It is known as lattice vibrations. The resistance of a metal to electrical current flow is due to the scattering of these free electrons by lattice vibrations. As the temperature increases, the amplitude of these lattice vibrations also increases, causing more frequent collisions between the free electrons and lattice vibrations. Consequently, the resistance of the metal increases with temperature. The dependence of resistivity ρ on temperature for a typical metal is shown in Figure 1(Blue curve).

The resistivity of a metal decreases with decreasing temperature and becomes very less close to the absolute zero temperature. This is because at low temperatures, the lattice vibrations are reduced, and the scattering of electrons is minimized. However, at $T=0$, the resistivity is not zero due to the presence of impurities in the metal. This residual resistivity is denoted by ρ_0 .

The temperature dependence of resistivity can be expressed by the Matthiessen's rule, which states that the total resistivity of a metal is the sum of its residual resistivity and its temperature-dependent part, as expressed in Equation 1.

$$\rho = \rho_0 + \rho(T) \text{ --- (1)}$$

Here, ρ is the resistivity of the given metal, ρ_0 is the residual resistivity, and $\rho(T)$ is the temperature-dependent part of the resistivity.

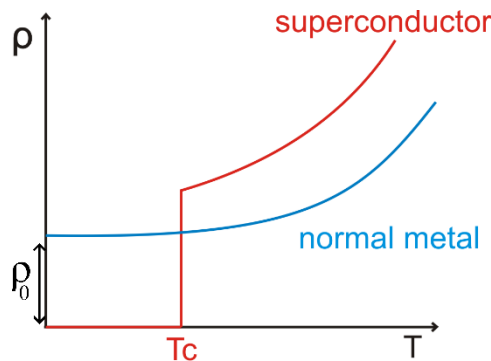


Figure 1. Temperature dependence of resistivity of a metal and a superconductor

Temperature dependence of resistivity:

Figure 1 (Red curve) shows the temperature dependence of resistivity ρ in a superconductor. In the non-superconducting state, ρ decreases with decreasing temperature, like a normal metal, until it reaches a critical temperature T_c . At T_c , ρ abruptly drops to zero, indicating the transition from the normal state to the superconducting state of the material. The critical temperature varies depending on the superconductor under investigation. For example, mercury becomes a superconductor with zero resistance at 4.2 K. Various superconductors, and their critical temperatures are listed in Table 1.

Table 1: Superconductors and their critical temperatures

Superconductor	Chemical Symbol	Critical Temperature (K)
Mercury	Hg	4.2
Lead	Pb	7.2
Niobium	Nb	9.3

Superconductor	Chemical Symbol	Critical Temperature (K)
Tin	Sn	3.7
Gallium	Ga	1.5
Indium antimonide	In Sb	1.9
Tungsten Carbide	WC	12.5
Magnesium Diboride	MgB2	39

Meissner effect

It was observed by Meissner and Ochsenfeld that, when a superconductor is cooled below its critical temperature (T_c), it undergoes a phase transition and enters a superconducting state. In this state, the superconductor exhibits perfect diamagnetism. It repels external magnetic field from its interior. This expulsion of magnetic flux is a characteristic feature of the Meissner effect.

Inside material, we know that

$$\vec{B} = \mu_0(\vec{H} + \vec{M}) \text{ ---- (2)}$$

Where, $\vec{B}$ represents the magnetic induction, $\vec{H}$ is the magnetic field intensity, $\vec{M}$ is the magnetization vector and $\mu_0 = 4\pi \times 10^{-7} \text{H/m}$, is the permeability of the free space. We can express magnetization vector $\vec{M}$ (the net dipole moment per unit volume) in the following form,

$$\vec{M} = \chi \vec{H}$$

where, χ is magnetic susceptibility. Substituting the value of magnetization vector in eq. 2, one gets the following expression:

$$\vec{B} = \mu_0(\vec{H} + \chi\vec{H}) = \mu_0(1 + \chi)\vec{H}$$

As we have observed for a superconducting sample, the magnetic field is zero inside the specimen, that is,

$$\begin{aligned} 0 &= \mu_0(\vec{H} + \chi\vec{H}) \\ \Rightarrow \chi &= -1 \end{aligned}$$

Thus, a superconductor is an ideal diamagnetic material.

For regular diamagnetic material,

$$|\chi| \ll 1$$

Magnetic flux density is given by

$$\vec{B} = \mu_0(\vec{H} + \vec{M})$$

For a superconducting material, $\vec{B} = \vec{0}$

$$\vec{0} = \mu_0(\vec{H} + \vec{M})$$

$$\vec{M} = -\vec{H}$$

$$\chi = -\frac{|\vec{M}|}{|\vec{H}|} = -1$$

In a superconducting pure metal, the magnetic flux is expelled from the sample, irrespective of its geometry and sequence in which the magnetic field is applied. The induction field decays inside

exponentially with distance from the surface of the sample. This characteristic decay-length is called the **penetration depth** (λ) and is generally estimated to be a few hundred angstroms (10^{-7} m).

The Meissner effect occurs due to the formation of Cooper pairs in the superconductor. Cooper pairs are composed of two electrons with opposite spins and momenta that form a bound state due to the attractive interaction mediated by lattice vibrations (phonons). These Cooper pairs collectively behave as a single entity, which extends throughout the entire superconductor (Mechanism of formation of such pair will be discussed afterwards).

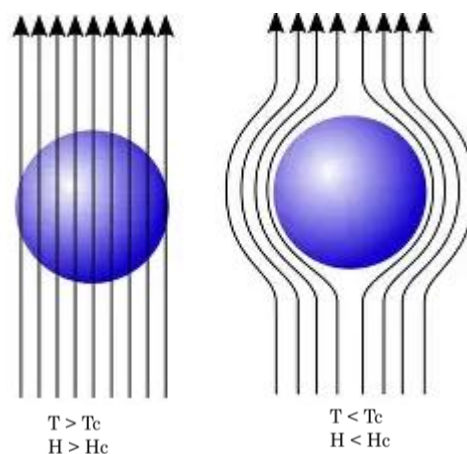


Figure 2. Meissner Effect

When an external magnetic field is applied to a superconductor, the magnetic flux tends to penetrate the material. However, the Cooper pairs actively respond to the magnetic field by forming super currents. These induced currents create a magnetic field that perfectly cancels out the external magnetic field within the bulk of the superconductor. As a result, the magnetic field is expelled from the interior of the superconductor, confining it to the exterior regions as shown in figure 2. The figure shows that above the critical temperature (T_c) or/and above a certain critical magnetic field (H_c), as the superconductivity is destroyed, the material behaves as a normal material allowing field lines through it.

According to Maxwell's equation,

$$\begin{aligned}\vec{\nabla} \times \vec{B} &= \mu_0 \vec{J}_s \\ \Rightarrow \vec{J}_s &= \vec{0}\end{aligned}$$

Thus, the supercurrent density is zero in the interior of a superconductor. Surface currents, however, can exist in a superconductor.

In some superconductors the Meissner effect is limited to specific magnetic field ranges which will be discussed in detail later.

Critical Current

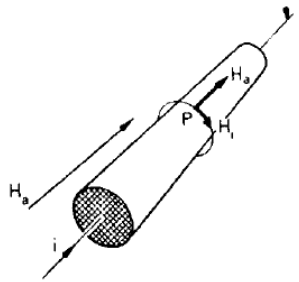


Figure 3. A superconducting current carrying wire in an applied magnetic field of H_a

Consider a current of i ampere is applied to a superconducting wire as shown in figure 3. The application of the current induces a magnetic field H_i . When the vector summation of this H_i and the applied field H_a crosses critical field H_c , the material no longer acts as a superconductor. This is called Silsbee effect.

Let us consider that the applied field is zero. The superconducting wire has a cylindrical cross-section of radius R . The current is flowing parallel to axis of the wire. The current is uniformly distributed over its cross-section. We will apply Ampere's circuital law to obtain the magnetic field generated at the surface of the wire due to this current. The law says: *The line integral of the magnetic field $\vec{B}$ around a closed loop is equal to the permeability of the medium μ times the total current i enclosed by the loop.*

We will consider now a circular loop of radius R around the wire which encloses a current i .

$$\oint \vec{B} \cdot d\vec{l} = \mu i$$

$\therefore B \oint dl = \mu i$ ($\vec{B}$ and $d\vec{l}$ are parallel vectors here (encircling the wire). Also, due to the symmetry of the wire, B is constant at all points on the circular loop, i.e., B is function of radial distance only)

$$B = \frac{\mu i}{2\pi R} \text{ as } \oint dl = \text{The circumference of the loop} = 2\pi R$$

$$\text{Now, } \vec{B} = \mu \vec{H}$$

$$\therefore H = \frac{i}{2\pi R}$$

So, the critical current required to destroy the superconducting property is given by

$$i_c = 2\pi R H_c \text{ (when } H_a=0\text{)}$$

Example 1. The critical magnetic field at 5 K is $2 \times 10^3 \text{ A/m}$ in a superconducting wire of radius 0.02 m. Find the value of critical current.

Solution.

$$I_c = 2\pi R H_c = 2 \times 3.14 \times 0.02 \times 2 \times 10^3 \text{ A/m} = \mathbf{251.4 \text{ A}}$$

Types of superconductors

Superconductors can be categorized as Type 1 or Type 2 superconductors based on their magnetization properties when subjected to an external magnetic field. Type 1 superconductors are also called *Soft superconductors*, while Type 2 superconductors are known as *Hard superconductors*.

Type-1 Superconductors

In soft superconductors, the magnetic field gets totally expelled from the interior of the material, below a certain critical magnetic field H_c . At the critical magnetic field, there is an abrupt loss of superconductivity. A plot of magnetization versus magnetic field for a Type-1 or soft superconductor is shown in figure 4.

The magnetic field can penetrate the material above the critical field H_c . The material above H_c is said to be in its normal state. This type of superconductor demonstrates complete Meissner effect at magnetic fields below H_c where, it becomes an ideal diamagnetic material. Lead, tin and mercury are common examples of soft superconductors. The transition to the superconducting state at H_c is

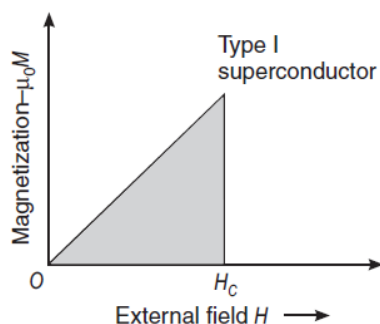


Figure 4. A plot of magnetization versus magnetic field

reversible. Most superconducting metals exhibit Type -1 superconductivity at very low H_c values (around 10^{-1} T).

Type-2 superconductors

Type-2 superconductors show two critical magnetic fields: H_{c1} and H_{c2} . Typical magnetization curves for hard superconductors are shown in figure 5.

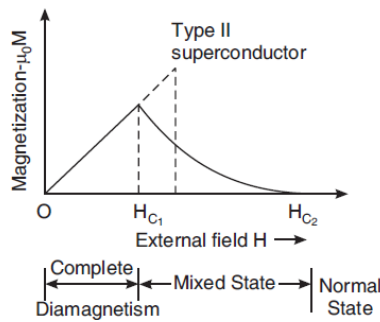


Figure 5. A plot of magnetization versus two critical magnetic fields

Till the first critical field H_{C1} , the material is superconducting and exhibits ideal diamagnetic behaviour. Magnetic flux is expelled completely from within the material. Between the two critical fields H_{C1} and H_{C2} the material exists in a mixed state (Figure 6). The material still demonstrates superconducting property, but exclusion of flux from within the specimen is partial. The bulk of the material is diamagnetic, the flux due to the applied field being opposed by a diamagnetic surface current which circulates around the perimeter of the specimen. This diamagnetic material is threaded by normal cores lying parallel to the applied magnetic field, and within each core is magnetic flux having the same direction as that of the applied magnetic field. These cores are called the vortices. Each vortex contains circulating supercurrent in a direction to support the flux penetration and hence the name 'vortex'. Vortices are normal region inside which the super electron density vanishes. As vortices contain parallel magnetic field lines, they repel each other like bar magnets and arrange themselves in form of a periodic lattice. This is known as vortex lattice. When magnetic field increases above H_{C1} , the vortex density keeps on increasing. At H_{C2} , the vortex density is such that they tend to touch each other transitioning the material into the normal state.

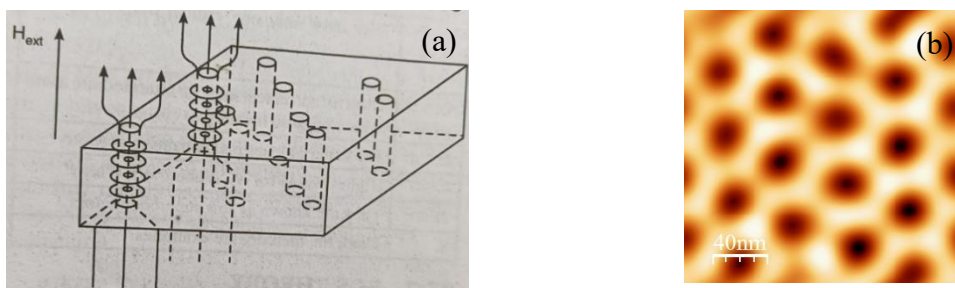


Figure 6. (a) Schematic of a Type 2 Superconductors with Vortices (b) Real image of a vortex lattice formed in niobium nitride (NbN) thin film. The image shows the conductance map of the top face of the film obtained by scanning tunneling microscope. The darker region has lower conductance representing normal vortex core while the lighter region has higher conductance indicating superconducting region. The picture demonstrates a hexagonal vortex lattice.

Alloys or transition metals with high electrical resistivity in the normal state generally demonstrate Type-2 superconductivity. The values of H_{c2} are 100 times more than the critical field values of Type-1 superconductors. H_{c2} values even up to 50 T have been obtained in some materials. As the magnetic field is increased above H_{c1} , magnetization observed in hard superconductors reduces gradually, whereas in soft superconductors, magnetization vanishes abruptly at the critical magnetic field. Hard superconductors are used as magnetizing coils to obtain high magnetic fields (10 T or higher).

Temperature dependence of critical field

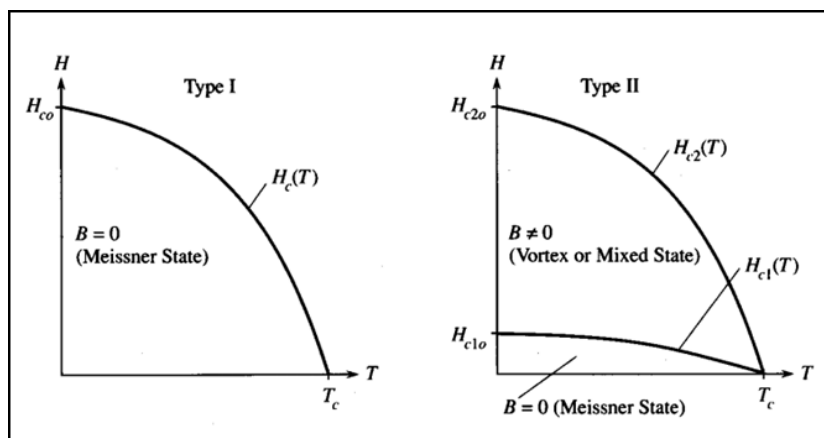


Figure 7. The temperature dependence of the critical field

The magnetic field plays a significant role in superconductivity. Superconductivity disappears at magnetic fields greater than a critical field H_c . The temperature dependence of this critical field is given by

$$H_c = H_0 \left[1 - \frac{T^2}{T_c^2} \right]$$

where, H_0 is the field required to destroy the superconducting property at 0 K, H_c is the minimum field required to destroy the superconducting property at T K and T_c the transition temperature of the material at zero field.

This temperature dependence of the critical field is given by Tuyn's law. Variation of the magnetic field with temperature is shown in figure 7.

Example 2. Calculate the critical current for a wire of lead having a diameter of 1 mm at 4.2 K. The critical temperature for lead is 7.18 K and $H_c(0) = 6.5 \times 10^4$ A/m.

Solution.

$$H_c(T) = H_0(0) \left[1 - \left(\frac{T}{T_c} \right)^2 \right] = 6.5 \times 10^4 \left[1 - \left(\frac{4.2}{7.18} \right)^2 \right] = 4.28 \times 10^4 \text{ A/m}$$

The critical current $I_C = 2\pi r H_C = \pi d H_C = 1 \times 3.14 \times 10^{-3} \times 4.28 \times 10^4 = \mathbf{134.5 \text{ A}}$

Example 3. The transition temperature for Pb is 7.2 K. However, at 5 K it loses the superconducting property if subjected to magnetic field of $3.3 \times 10^4 \text{ A/m}$. Find the maximum value of H which will allow the metal to retain its superconductivity at 0 K.

Solution.

$$H_0 = H_c / \left[1 - \frac{T^2}{T_c^2} \right] = \frac{3.3 \times 10^4}{1 - (25/51.28)} = 6.37 \times 10^4 \text{ A/m}$$

Example 4. The critical field of niobium is $1 \times 10^5 \text{ A/m}$ at 8 K and $2 \times 10^5 \text{ A/m}$ at 0 K. Calculate the transition temperature of the element.

Solution.

$$T_c = \frac{T}{\left[1 - \frac{H_c(T)}{H_c(0)} \right]^{1/2}} = \frac{8 \text{ K}}{\left[1 - \frac{1 \times 10^5 \text{ A/m}}{2 \times 10^5 \text{ A/m}} \right]^{1/2}} = 11.3 \text{ K}$$

Example 5. The transition temperature for lead is 7.26 K. The maximum critical field for the material is $8 \times 10^5 \text{ A/m}$. Lead has to be used as a superconductor subjected to a magnetic field of $4 \times 10^4 \text{ A/m}$. What precaution will have to be taken?

Solution.

$$T = T_c \left[1 - \frac{H_c(T)}{H_c(0)} \right]^{\frac{1}{2}} = 7.26 \text{ K} \left[1 - \frac{4 \times 10^4 \text{ A/m}}{8 \times 10^5 \text{ A/m}} \right]^{\frac{1}{2}} = 7.08 \text{ K}$$

Therefore, the temperature of the metal should be held below 7.08 K.

BCS theory

The BCS theory, a microscopic theory proposed by Bardeen, Cooper and Schrieffer in 1957, is based on the formation of *Cooper pairs*.

A pair of electrons formed by the interaction between the electrons with opposite spin and momenta in a phonon field is called a Cooper pair.

The basic idea that even a very weak attractive potential can result in bound pairs of electrons, known as Cooper pairs, was first formulated by Cooper in 1956. Later, BCS theory could successfully

extend the concept for a multiparticle wavefunction representing the superconducting state. This theory explains most of the phenomenon associated with superconductivity. During the flow of current in a superconductor, when an electron approaches a positive ion of the metal lattice, there is Coulomb attraction between the electron and the lattice ion. This produces a distortion in the lattice i.e., the positive ion gets displaced from its mean position. The distortion gives rise to a **phonon**. Phonons are quanta of lattice vibrations. Smaller the mass of the positive ion core, the greater the distortion will be. This interaction called the electron-phonon interaction leads to a scattering of the electron and causes electrical resistivity. The distortion causes an increase in the density of ions in the region of distortion. The higher density of ions in the distorted regions attracts, in turn, another electron. Thus, a free electron exerts an attractive force on another electron through phonons. This is how the Cooper pairs are formed via lattice mediation overcoming the coulomb repulsion. Suppose an electron of wave vector $\vec{K}$ emits a virtual phonon $\vec{q}$ which is absorbed by another electron having wave vector $\vec{K}'$. Thus, $\vec{K}$ is scattered as $\vec{K} - \vec{q}$ and $\vec{K}'$ as $\vec{K}' + \vec{q}$ as shown in figure 8. The resulting electron-electron interaction depends on the strength of the attractive potential due to the phonons (V_{ph}). As BCS theory considers pairing of electrons with equal and opposite momenta and opposite spins ($\vec{K}_{\uparrow}$ and $-\vec{K}_{\downarrow}$), the Cooper pair has a total momentum of zero, and total spin of zero. As a result, the electron pairs in a superconductor are bosons.

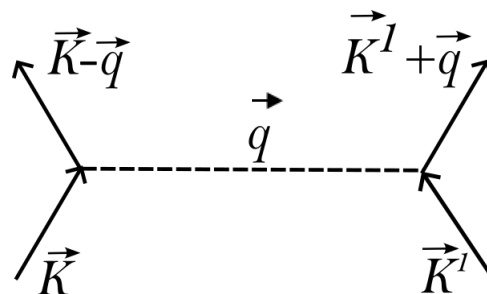


Figure 8. Conservation of momentum for electron-phonon interaction

At higher temperatures, i.e., above the critical temperature, the attractive force is small as the lifetime of the phonons decreases though number of phonons becomes more. Also, thermal energy is too large to let the pairing of electrons happen.

At lower temperatures, i.e., below the critical temperature the apparent force of attraction between electrons exceeds the Coulomb force of repulsion between them and the electrons form the bound pairs.

Concept of a phase coherent state according to BCS theory

In a typical superconductor, the volume of a given pair encompasses as many as 10^6 other pairs. This dense cloud of Cooper pairs form a collective state where strong correlations arise among the motions of all pairs because of which they drift cooperatively through the material. Let us consider that the wavefunction of a Cooper pair is given by $\psi = Ae^{i\theta}$, where A is the amplitude which is proportional to the energy to break the Cooper pair. θ is the phase part of the wavefunction. According to BCS theory, all the Cooper pairs are phase coherent i.e. the phase for all the pairs are same. Thus, superconducting state is an ordered state of the Cooper pairs. Either they are at rest, or if the superconductor carries a current, they drift with identical velocity. Since the density of Cooper pairs is quite high, even large currents require only a small velocity. They do not collide as there are no vacant states available for them to go to after scattering. Because of this, the substance possesses infinite electrical conductivity.

When the electrons flow in the form of Cooper pairs in materials, they do not encounter any scattering and thereby resistance vanishes, or conductivity becomes infinity which is known as superconductivity.

Limitations of BCS Theory:

1. BCS theory is only applicable to low temperature, clean and three-dimensional superconductors. It does not explain high-temperature superconductivity and low-dimensional superconductivity.
2. It assumes that the electron-phonon interaction is weak. Weak-coupling materials like Aluminum and mercury fit the BCS prediction well. For strong coupling cases, like MgB_2 , BCS theory is not suitable.
3. It only assumes Cooper pairs formed by bound electrons of opposite spins. It does not consider other pairings like pairs formed by electrons having same spins.
4. The theory does not predict the probability of superconductivity.
5. Cases which require time-dependent or non-equilibrium models are not explained in BCS theory.
6. Superconductivity, which is not mediated by phonons, is not captured in BCS theory.

Two-fluid model:

Material in superconducting state is made up of two types of charge carriers. One part is made up of cooper pairs and the other part is composed of normal electrons. So, the total carrier density at any temperature below T_c is, $n = n_n + n_s$, where n_n is the normal electron density and n_s is the super electron density. The fractions of n_s and n_n vary depending on the temperature. Normal electrons can undergo scattering, but super electrons do not undergo scattering and have only inductive effects during current conduction. This construction can be modelled as an electrical circuit where a resistance representing n_n is in parallel to an inductor representing n_s (figure 9). When we send dc current through a superconductor, as the frequency of the current is zero, the inductive path acts as a short. Thus, current always chooses the inductive path leading to no voltage drop or dissipation. **This is known as two-fluid model.**

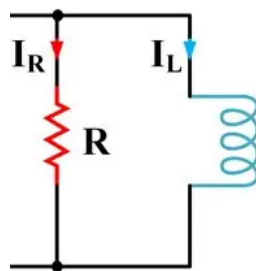


Figure 9 Representation of the two-fluid model as an electrical circuit

Quantum tunnelling:

The word ‘tunnelling’ came into use with the advent of quantum mechanics; it was observed that electrons possess the unique property of sneaking through nano-sized barriers to contribute to the minority carriers in semiconductors. The probability of tunnelling is small; the transmission coefficient is of the order of 10^{-1} to 10^{-3} depending on the barrier height, barrier width and the energy of the particle. Then a similar question came to the mind of the scientists dealing with superconductors. *Do the Cooper pairs have this unique property of tunnelling?* In 1962, B.D Josephson predicted several remarkable phenomena about superconductivity. When two superconductors are joined together by a thin layer of oxide, Josephson explored the probability of tunneling for the newly found cohesive pair of electrons that is the Cooper pairs, through that thin insulating barrier. Cooper pair tunnelling finds many applications related to superconductors like magnetometry, quantum computing etc.

DC Josephson Effect

When we have a normal material-superconductor boundary, the super electron density gradually falls across the junction from a constant non-zero value in the superconducting part to zero in the normal part. The **coherence length** ξ is a fundamental length scale in a superconductor that characterizes the spatial extent over which the superconducting wavefunction ψ can vary appreciably (Figure 10).

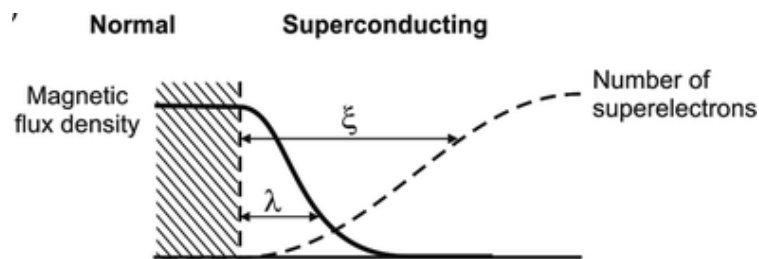


Figure 10 Variation of Magnetic flux density and Number of Cooper pairs across a normal material-superconductor boundary. Characteristic lengths (coherence length ξ and penetration depth λ) are indicated in the diagram.

Physically, it represents:

- The order of magnitude of the size of a Cooper pair (the spatial correlation between the two paired electrons)
- The minimum length scale over which the superconducting state can be "healed" if disturbed (e.g., near an interface or defect)

An insulating material of thickness nearly 1 to 2 nm (much less than the coherence length of the material) sandwiched between two different superconducting materials is known as Josephson device (figure 10). In a Josephson device, a dc super current is found to flow from one superconducting electrode to the other. This phenomenon is said to be DC Josephson effect. The dc current flowing through the device is equal to

$$I = i_c \sin \delta$$

Where, δ is the phase difference between the wave functions of super electrons of the two superconducting electrodes and i_c is a constant for that junction. It represents the maximum current through the insulator which can flow without any voltage drop across the junction (critical current of the junction).

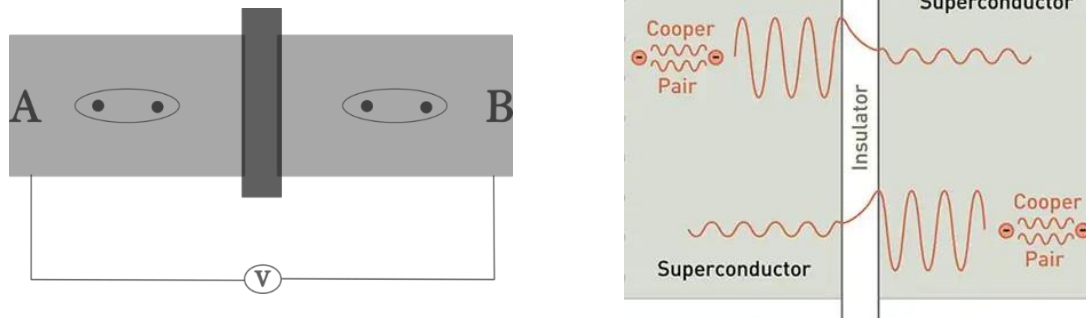


Figure 11. Josephson device and tunneling through the Josephson Junction

The cooper pairs present in a superconducting material are in the same phase. Whenever a Josephson junction is formed by sandwiching an insulator in between two pieces of superconductors, the superconducting electrons present in the two different superconducting electrodes need not be in the same phase. When the width of the insulating barrier is less than the coherence length of the superconducting electrodes, tunnelling of cooper pairs occurs. Wavefunction from one side of the barrier extends to the other side through the barrier even though it decays while crossing the barrier (figure 11). This tunnelling results in a net supercurrent or Josephson current depending on the phase difference of the wavefunctions of the cooper pairs of the electrodes. The direction of the current is governed by the sign of the phase difference. The supercurrent can be sustained by adding a current source to the circuit. The voltmeter in figure 11 will show zero voltage drop if the current sent is below the critical current of the junction.

AC Josephson effect

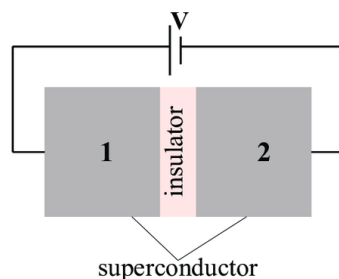


Figure 12 Ac Josephson effect

Consider a dc voltage V applied to a Josephson device (figure 12). The phase difference δ evolves with time at a rate, $\frac{d\delta}{dt} = \frac{2eV}{\hbar}$. Thus, the current through the junction becomes an alternating current. The Josephson current is given by

$$I = i_c \sin\left(\frac{2eV}{\hbar}t\right) = i_c \sin(\omega t)$$

Where $\omega = \frac{2eV}{\hbar}$ is the angular frequency of the ac signal. The application of a potential difference makes the Josephson current time dependent. This phenomenon is known as AC Josephson effect.

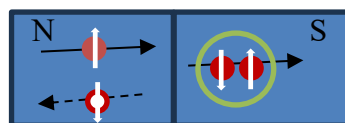
The junction can emit electromagnetic radiation because of this oscillating current. The photon energy of emission at the junction is $h\nu = qV = 2eV$.

$$\therefore \nu = \frac{qV}{h} = \frac{2eV}{h} = \frac{2 \times 1.6 \times 10^{-19} \times V}{6.625 \times 10^{-34}} = 483.5 \times 10^{12} \text{ V Hz}.$$

For an applied voltage of $1\mu\text{V}$, the frequency of the ac signal is $\nu = 483.5 \times 10^{12} \text{ V} = 483.5 \times 10^{12} \times 10^{-6} = 483.5 \text{ M Hz}$.

Josephson device can also have superconductor - normal material - superconductor (S-N-S) architecture where we have two superconductor-normal material interfaces. A normal electron after being incident at an NS interface cannot directly enter the superconductor. It pairs up with another electron of opposite spin forming a cooper pair and then proceeds through the superconductor. The incident electron has less energy than the superconducting gap of the superconductor. In this process, we obtain a retroreflected hole in the normal material with opposite spin and velocity of the incident electron. This retroreflection ensures conservation of charge and momentum. This phenomenon is known as Andreev reflection (figure 13). For S-N-S junctions, supercurrent can be formed via multiple Andreev reflections (also known as Andreev tunneling). But this current is non-sinusoidal.

Figure 13. Andreev reflection



Example 6. A Josephson junction with a voltage difference of $650 \mu\text{V}$ radiates electromagnetic radiation. Calculate its frequency.

Solution.

$$\nu = \frac{2eV}{h} \text{ Hz} = \frac{2(1.6 \times 10^{-19})(650 \times 10^{-6})}{6.625 \times 10^{-34}} = 3 \times 10^{11} \text{ Hz}$$

High- T_c Superconductors:

The temperature T_c below which a material shows superconductivity is an important criterion in deciding the application of a superconductor. The earliest known superconductors needed to be cooled to temperature close to 4 K (boiling temperature of helium) to display superconducting properties. Though such temperatures are achievable, cooling to such a low temperature involves enormous expenses.

Continuous research is being carried out to discover superconductors with higher T_c values. Such superconductors are collectively referred to as high- T_c superconductors (HTSCs). But the formal definition of a high- T_c superconductor is the superconductor which shows a T_c above 77 K (table 2).

Many modern high temperature superconductors have a complex unit structure with oxygen called Perovskite structure. Such cells are made of one atom of rare earth metal, two Barium atoms, three Copper atoms and seven oxygen atoms and such compounds are popularly referred to as 1-2-3 super conductors.

In 1986, K.A Muller and J.G Bednorz reported superconducting properties of Lanthanum, Barium and Copper oxides at temperatures higher than 30K. This was followed by the synthesis of Yttrium compounds by C.W Chu in 1987. These Yttrium compounds acquired the superconducting state with a T_c close to 90K. Several HTSCs have since been reported with $T_c > 30K$. The discovery of these HTSCs rekindled the interest in superconductivity from the application viewpoint.

Table 2: Examples of High T_c Superconductors and their critical temperatures

Material	$T_c(K)$
$LnBa_2Cu_3O_7$	90
$Bi_2(CaSr)_{n+1}Cu_nO_{2n+4}(n=1-3)$	90
$Tl_2Ca_{n-1}Cu_nO_{2n+4}(n=1-4)$	90-110

In 2019 it was discovered that lanthanum hydride (LaH_{10}) becomes a superconductor at 250 K under a pressure of 170 gigapascals. As of 2021, the superconductor with the highest transition temperature at ambient pressure was the cuprate of mercury, barium, and calcium, at around 133 K ($-140^\circ C$). Very recently in 2025, single crystal of $HgBa_2Ca_2Cu_3O_{8+\delta}$ (Hg-1223) has shown the highest T_c of 134 K at ambient pressure but the reason is still elusive.

These high temperature superconductors are not commercially viable mainly due to their inherent brittleness, difficulty in forming into wires and inability to carry high current densities. Once

these difficulties are overcome, such superconductors find potential applications in zero loss power transmission lines, super strong magnetic materials and as materials for levitating trains.

SQUID:

SQUID is an acronym for Superconducting Quantum Interference Device. It is used as a magnetometer. It is based on the principle of Josephson effect.

DC – SQUID:

Consider two Josephson junctions are connected in parallel as shown in figure 14. Let the current be applied through a common arm T1 and a magnetic field is applied perpendicularly to this device. The current is split into two components to flow through both the Josephson junctions. It is like the splitting of light into two coherent sources in Young's double slit experiment. The current leaving the Josephson junctions (I_a and I_b) are combined at the common arm T2. It is like the two split beams of light who are allowed to interfere with each other in Young's double slit experiment. The current flowing through the arm T2 is

$$I = I_a + I_b = 2i_c \sin(\delta) \cos \frac{e\varphi}{\hbar}$$

Where δ is the average phase difference across the junctions, i_c is the critical current of individual junctions (here assumed to be same for both the junctions) and φ is the total magnetic flux.

The maximum current will flow, when $\delta = 90^\circ$,

$$I_{max} = I_c = (I_a + I_b)_{max} = 2i_c \cos \frac{e\varphi}{\hbar}$$

I_c is critical for the SQUID device.

Substituting $\hbar = \frac{h}{2\pi}$, we get

$$I_c = 2i_c \cos \frac{2\pi e\varphi}{h} = 2i_c \cos \frac{\pi\varphi}{\varphi_o}$$

Where φ_o is known as flux quantum and

$$\varphi_o = \frac{h}{2e} = \frac{6.626 \times 10^{-34}}{2 \times 1.6 \times 10^{-19}} = 2.07 \times 10^{-15} \text{ Wb}$$

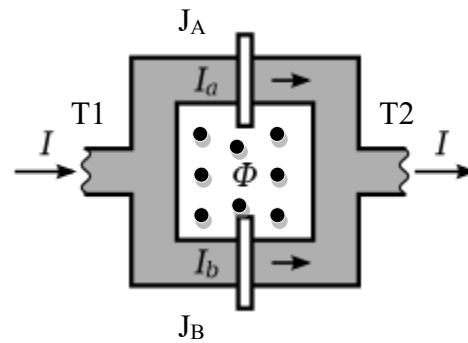


Figure 14 Schematic of a DC SQUID device

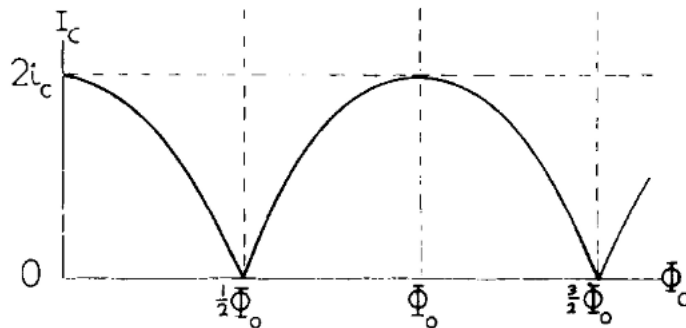


Figure 15 Variation of critical current in a SQUID as a function of applied magnetic field

So, the critical current of the SQUID is an oscillating function of applied magnetic field with a period of φ_0 (Figure 15).

Physical Origin of this periodicity:

In a superconductor, the macroscopic wavefunction describing Cooper pairs must be single valued. When a superconducting loop encloses magnetic flux, the phase of the wavefunction changes. To maintain single-valuedness of the wavefunction after traversing a full closed path around the superconducting loop, the phase shift must be an integer multiple of 2π . When we start from any point on a closed loop, the wavefunction $\psi_0 = Ae^{i\theta}$. After completing one full traversal of the loop, $\psi_{2\pi} = Ae^{i(\theta+2\pi)}$. To achieve single-valuedness, $\psi_0 = \psi_{2\pi}$. For n number of full traversals, $Ae^{i\theta} = Ae^{i(\theta+2\pi n)}$. This property of the wavefunction leads to the condition that the magnetic flux enclosed by any superconducting loop must be an integer multiple of flux quantum, φ_0 . This is known as **Flux Quantization**.

Usually, the critical current of a Josephson junction is much smaller than that of the individual superconducting electrodes. So, the circulating current in the SQUID is usually very small. However, the circulating current can introduce a significant phase difference across the weak links. As the magnetic field is varied, the circulating supercurrent adjusts itself to maintain integer multiple of φ_0 through the SQUID loop. This leads to the oscillation of the critical current with period of φ_0 .

The critical current profile as a function of applied field helps us to measure the applied magnetic flux which is of the order of φ_0 . Hence, it is possible to measure very small magnetic flux using these arrangements.

DC SQUIDS are used in

- ✓ Measurement of magnetic fields as small as 10^{-21} T produced by biological currents in human heart and brain can be measured.
- ✓ Geophysical measurements (connected with rocks, seismic waves etc)
- ✓ Nondestructive testing (testing the material without damaging it)
- ✓ Readout of qubits
- ✓ Very precise and accurate measurement of voltage.

RF-SQUID or AC SQUID

The RF (Radio Frequency) SQUID is a one-junction SQUID loop and is used as a magnetic field detector. Although it is less sensitive than the DC SQUID, it is cheaper and easier to manufacture and is therefore more commonly used (figure 16).

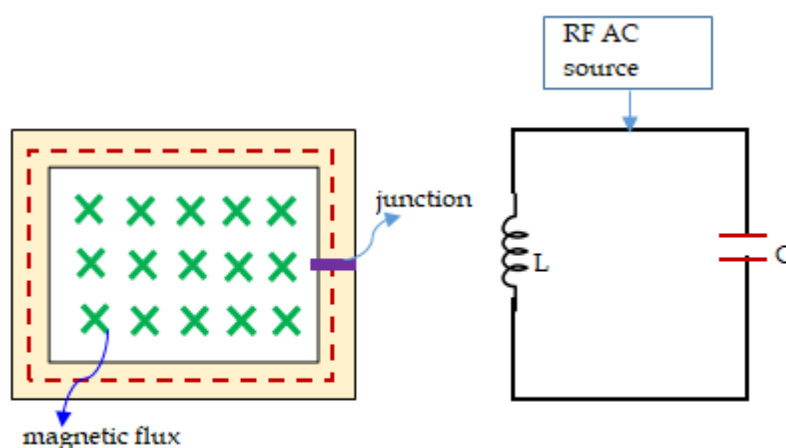


Figure 16 Schematic of an RF SQUID device

RF SQUID loop is usually placed near an LC circuit which is connected to RF AC source. The loop is immersed in a magnetic field whose flux is Φ (to be measured). Now pass an oscillating current (I) through LC circuit from the RF source. It induces magnetic flux Φ_{RF} . This flux is coupled with the loop. The total external flux is

$$\Phi_{ex} = \Phi + \Phi_{RF}$$

The loop is linked to LC circuit through mutual inductance. By chance if the flux in the loop (Φ) changes, there will be corresponding changes in Φ_{ex} . Any changes in the total flux Φ_{ex} will induce emf (according to Faraday's law) in the LC circuit and hence voltage (V) across LC changes. By measuring the change in V one can measure the external magnetic flux (Φ) and its variation with respect to time.

RF-SQUIDS are used in

- ✓ Bio magnetism (i.e., to measure magnetic field produced by different organs of our body)
- ✓ Geophysical measurements (connected with rocks, seismic waves etc.)
- ✓ Nondestructive testing (testing the material without damaging it)

Questions:

1. The dc resistance of a superconductor is practically zero. What about its ac resistance?
2. What are Cooper pairs?
3. How is Josephson tunnelling different from single particle tunnelling?
4. A superconducting wire and a copper wire are connected in parallel. Does the copper wire carry current when a potential difference is applied?
5. What is the significance of critical temperature, critical magnetic field and critical current density for superconductors?
6. What is Meissner effect?
7. Compare the dependence of resistance on temperature of a superconductor with that of a normal conductor.
8. What do you mean by "perfect diamagnetism" in the context of a superconductor?
9. Describe how Cooper pairs are formed and explain the salient features of superconductivity?
10. Explain the term high temperature superconductivity. Give the various applications of superconductors.
11. What are type I and type II superconductors? Explain key points of B.C.S theory.

12. Explain ac and dc Josephson effect.
13. Describe the principle of working of a DC SQUID and mention its applications.
14. Discuss the principle and working of an RF SQUID.